ENHANCE

Is Your Carbon Claim Physically Defensible?

FIELD CASE SIGNAL A - TIST KENYA • AGROFORESTRY

Growth-based claim • Comparable pathway • VCS-verified • Exeter University Independent assessment

KENYA • -0.5°
Agroforestry • 18.278ha

62,278

tCO_2e/yr • GeoCarb Core estimate

+3.8%

vs. Independent proxy (~60,000tCO_2e/yr)

✓ PASS

LAOS • +18°
REDD+ / Avoided deforestation • 13.5M ha

-59.7%

Declared: 3,669,638 tCO_2e/yr
Screening-adjusted: 1,478,405 tCO_2e/yr

× DISTORTION

Claim materially diverges from defensible screening-adjusted result.

BENCHMARK • NASA MODIS MOD17A3 NPP

27 latitude bands • 7 vegetation classes • independent satellite-derived productivity patterns

0.91	0.9883	27
R^2 • Aggregated vegetation categories	R^2 • Wildflower-specific benchmark	Latitude bands tested

SCREENING OUTPUT SIGNALS

PASS
Claim appears physically plausible within GeoCarb's declared screening boundary.

WARNING
Claim may be plausible but requires additional evidence or methodological clarification.

DISTORTION
Claim materially diverges from screening reference. Review required before capital.

> **tCO_2e is a unit — not a description of a carbon process.**
> *Classification before comparison. Diagnosis before capital.*
> *GeoCarb™ v2.0 • ENHANCE Institute • June 2026 • ENH-GC-RPT-001-2026*

Carbon Claim Screening for REDD+ and NbS
Public Research Report • ENHANCE Institute

Full report" SSRN Working Paper • April 2026
contact@enhance-institute.org

Full Analysis: GeoCarb™ v2.0 Report

ENHANCE

GeoCarb™

Carbon Claim Screening for REDD+ and NbS

A Physical Plausibility and Comparability Screen for NbS Claims Before Funding, Purchase, or Certification

Structure determines outcome.
Design determines survival.

ENHANCE Institute

June 2026

Enhancing the Systems that sustain our world

Carbon Claim Screening for REDD+ and NbS: A Physical Plausibility and Comparability Screen for NbS Claims Before Funding, Purchase, or Certification

GeoCarb™ v2.0 is a carbon claim comparability classifier and physical plausibility screening engine for nature-based solution projects at the design stage.

It operates through a classification-first architecture. Before any physical comparison is made, GeoCarb™ v2.0 identifies the type and boundary of the carbon claim being made, then routes the claim to the appropriate screening pathway.

Model outputs are generated through rule-based computational procedures. The public report discloses the model's screening philosophy, claim-classification approach, architectural boundaries, and illustrative field case signals. Internal calibration logic, routing thresholds, parameter values, risk modifiers, component-level diagnostic rules, and proprietary system integrations are not disclosed herein and remain protected intellectual property of ENHANCE Institute.

This architecture ensures that GeoCarb™ outputs are generated through deterministic structured procedures rather than subjective judgment — while keeping the proprietary engine protected.

The core academic paper describing GeoCarb™'s foundational methodology is available via SSRN (April 2026).

About ENHANCE

ENHANCE Institute is the research division of ENHANCE Co., Ltd. It designs sustainable policies, projects, and systems based on deterministic architectural principles.

It develops integrated architectures that ensure climate interventions are structurally viable, physically grounded, and capable of delivering real-world outcomes.

To support this, ENHANCE operates proprietary evaluation frameworks — including VERA™, ToC: f(x)™, C-FAIR™, VORTA™, ESF™, GeoCarb™, and PathFinder-GOV™ — which verify the structural integrity, risk allocation, carbon claim plausibility, value formation, and pre-submission alignment of designed systems.

enhance-institute.org | contact@enhance-institute.org

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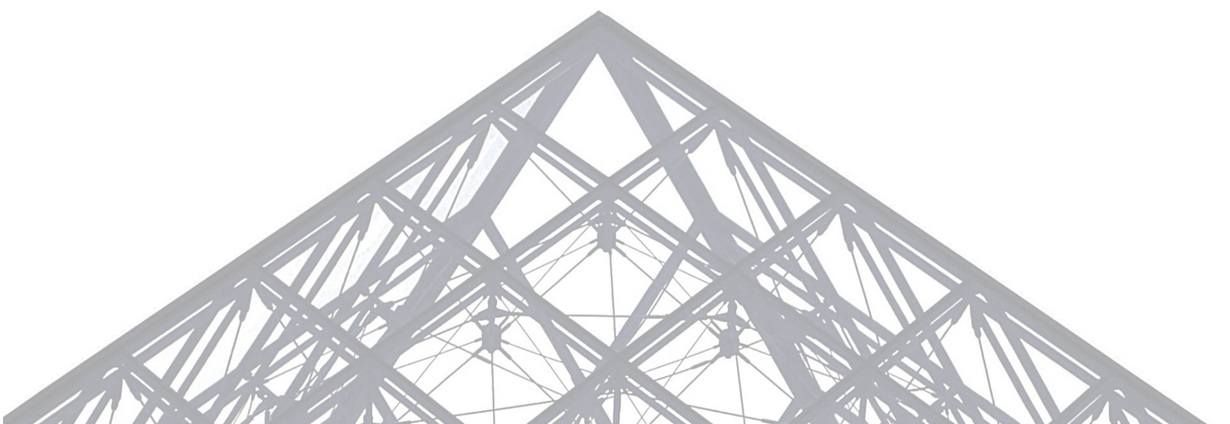
THE GAP THAT GeoCarb™ FILLS

LATITUDE-BASED ABSORPTION POTENTIAL

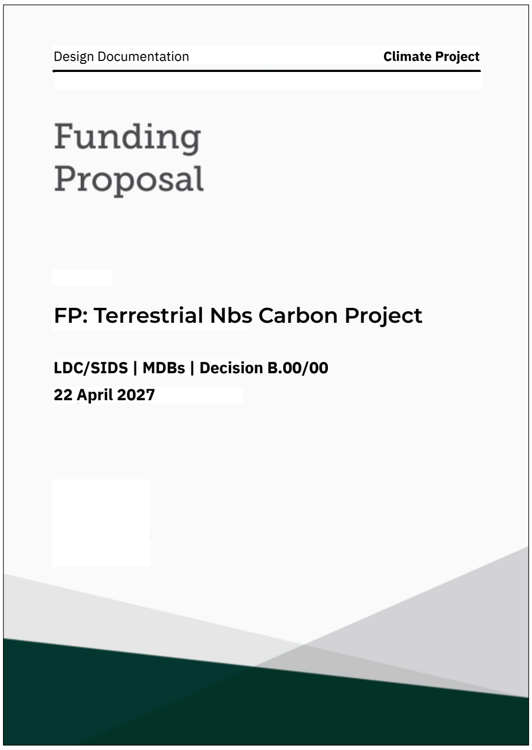
GeoCarb™ v2.0 — Public Screening Output



Contents

Project Design / Claim Draft

GeoCarb™


Design Documentation | Climate Project

Funding Proposal

FP: Terrestrial Nbs Carbon Project

LDC/SIDS | MDBs | Decision B.00/00
22 April 2027

Approval

Verra/GS Validation & Verification Pathway

Before the Claim Enters Finance

GeoCarb™ operates before certification.

It sits at the design and proposal stage, where a nature-based carbon claim first becomes a financial claim.

It does not replace Verra, Gold Standard, field MRV, or ecological survey.

It asks two prior questions:

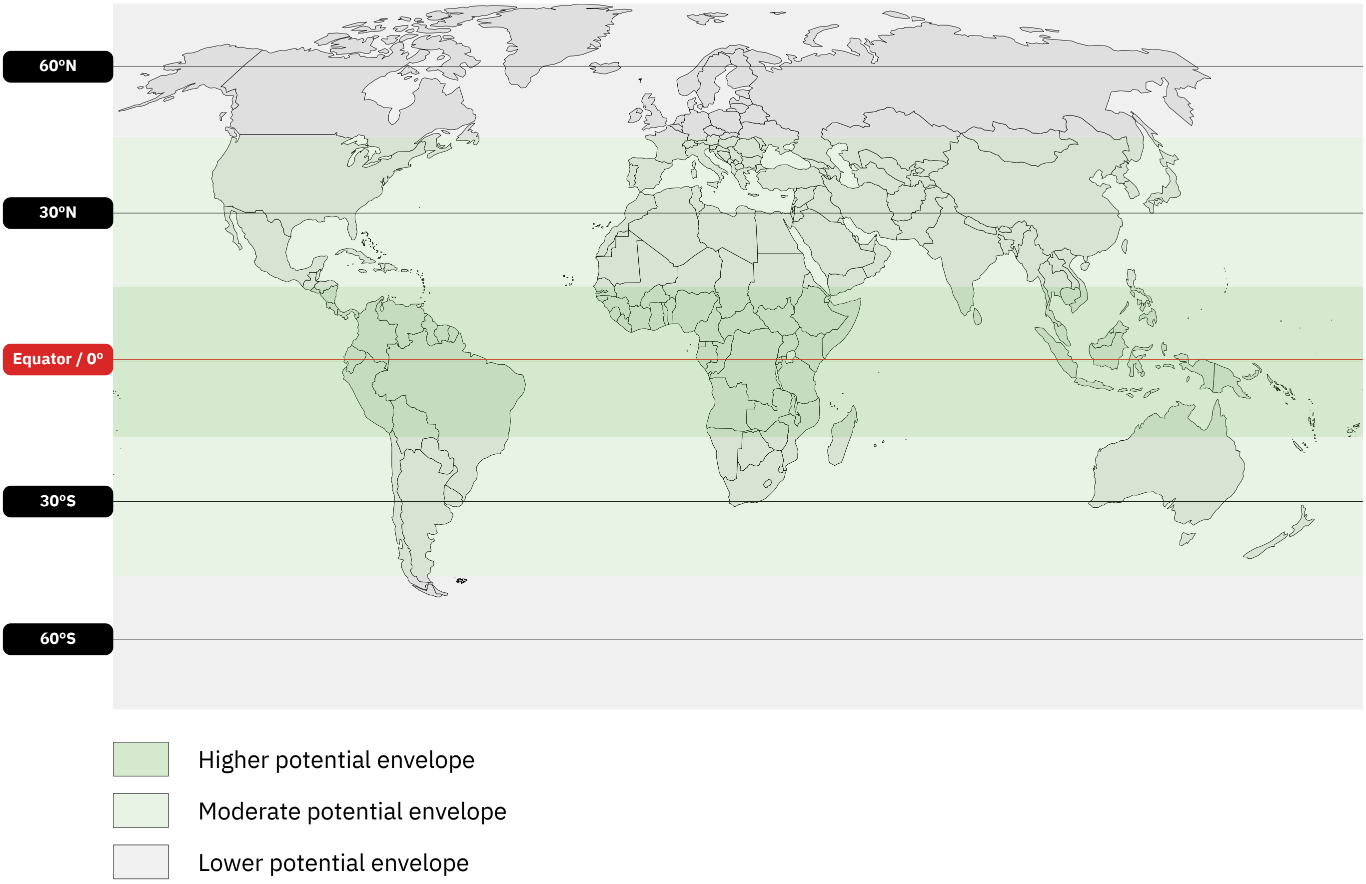
What kind of carbon claim is this — avoided emission, additional removal, or a mixture?

And is the claimed carbon outcome physically plausible at this location, for this vegetation, at this scale?

If the answer to either question is not established, the claim should not become the basis for funding, purchase, or certification preparation.

GeoCarb™ before capital. Not instead of certification.

Claim type first. Physical plausibility second. One answer before funding.



Latitude Sets the Physical Screen — Not the Whole Model

Carbon absorption is not spatially uniform.

At the design stage, latitude provides the first physical constraint: it shapes solar availability, growing-season structure, and the broad productivity envelope within which terrestrial vegetation can absorb carbon.

GeoCarb™ v2.0 does not treat latitude as the only variable, nor as the starting point of evaluation.

Before latitude is applied, GeoCarb™ v2.0 determines what kind of carbon claim is being made. Only claims classified as Additional Removal or Removal Enhancement enter the latitude-based GeoCarb™ Core pathway directly.

Avoided-emission claims (REDD+, conservation) follow a separate decomposition pathway that does not rely on latitude-based absorption comparison.

This page illustrates the Core layer's physical envelope — one component of GeoCarb™ v2.0's multi-layer architecture.

> **A carbon claim should be tested against the physical absorption potential of its actual location before it becomes the basis for funding, purchase, or certification preparation. GeoCarb™ v2.0 tests more than that.**

The map is conceptual. It does not disclose GeoCarb™ calibration parameters, threshold values, or internal correction layers.

```
================================================================
  GeoCarb™ Public Screening Report
  GeoCarb™ v2.0 | ENHANCE Institute | 2026-05-26
================================================================

  Project Context
  Species            : agroforestry
  Area               : 18,278 ha
  Latitude           : -0.5° (S)
  Country            : KEN
  Project Type       : ARR

  Comparability Decision
  🟢 COMPARABLE — Direct comparison with GeoCarb™ Core applicable
  Routing: GeoCarb Core + GSM comparison

  Growth Stage
  Stage    : [redacted]
  Source   : age (8yr)
  Note     : [redacted]

  Screening Result
  GeoCarb™ Core potential    : 62,277.7 tCO2e/yr
  GSM-adjusted C_reference   : 17,437.8 tCO2e/yr [redacted]
  Claimed                    : 50,000.0 tCO2e/yr
  Deviation (GSM basis)      : [redacted]
  tco2_final                 : [redacted] tCO2e/yr [REPLACED]
  Flag                       : 🔴 DISTORTION
  Review status              : Independent verification recommended

  Risk Context
  Country-context risk adjustment applied.

  Interpretation
  The project claim (50,000 tCO2e/yr) is lower than the full
  GeoCarb™ established potential (62,278 tCO2e/yr), but
  significantly exceeds the current growth-stage-adjusted
  estimate.
  Because the project is at GS-EARLY stage, GeoCarb™ applies GSM =
  0.28 before comparison. On this basis, the claim exceeds the
  current-stage defensible estimate by 65.1%. VERA RA1 signal
  triggered.
  [redacted]

  ------------------------------------------------------------
  This report provides a high-level GeoCarb™ screening result.
  Detailed model parameters, calibration values, and internal thres
  Note: 'Screening-adjusted' denotes GeoCarb™ internal screening
        result only. Not a certified or issued carbon credit.
================================================================
```

Public-output illustration only.
Screening-adjusted results are not certified or issued carbon credits.

The screen shown here is an illustrative GeoCarb™ v2.0 CLI output generated in public-output mode.

Public mode is designed to show the minimum level of information needed for external readers to understand the screening result without exposing the protected model engine. It presents the project context, the claim's comparability status, the screening result, the review signal, and a high-level interpretation of what the output means.

This mode allows readers to see that GeoCarb™ is an operating screening engine, not only a conceptual framework. The model receives a declared nature-based carbon claim, classifies whether the claim can be screened within GeoCarb™'s declared boundary, and reports whether the claim appears plausible, requires review, or materially diverges from the screening reference.

Public mode does not disclose internal calibration parameters, routing thresholds, sensitivity ranges, vegetation-specific coefficients, diagnostic equations, risk modifiers, or ENHANCE integration logic. Those elements remain protected.

The output should therefore be read as a public screening signal, not as a certified credit volume, legal determination, investment recommendation, or substitute for project-level MRV.

In short, public mode shows how GeoCarb™ reports a screening result.

It does not reveal how the protected engine is reconstructed.

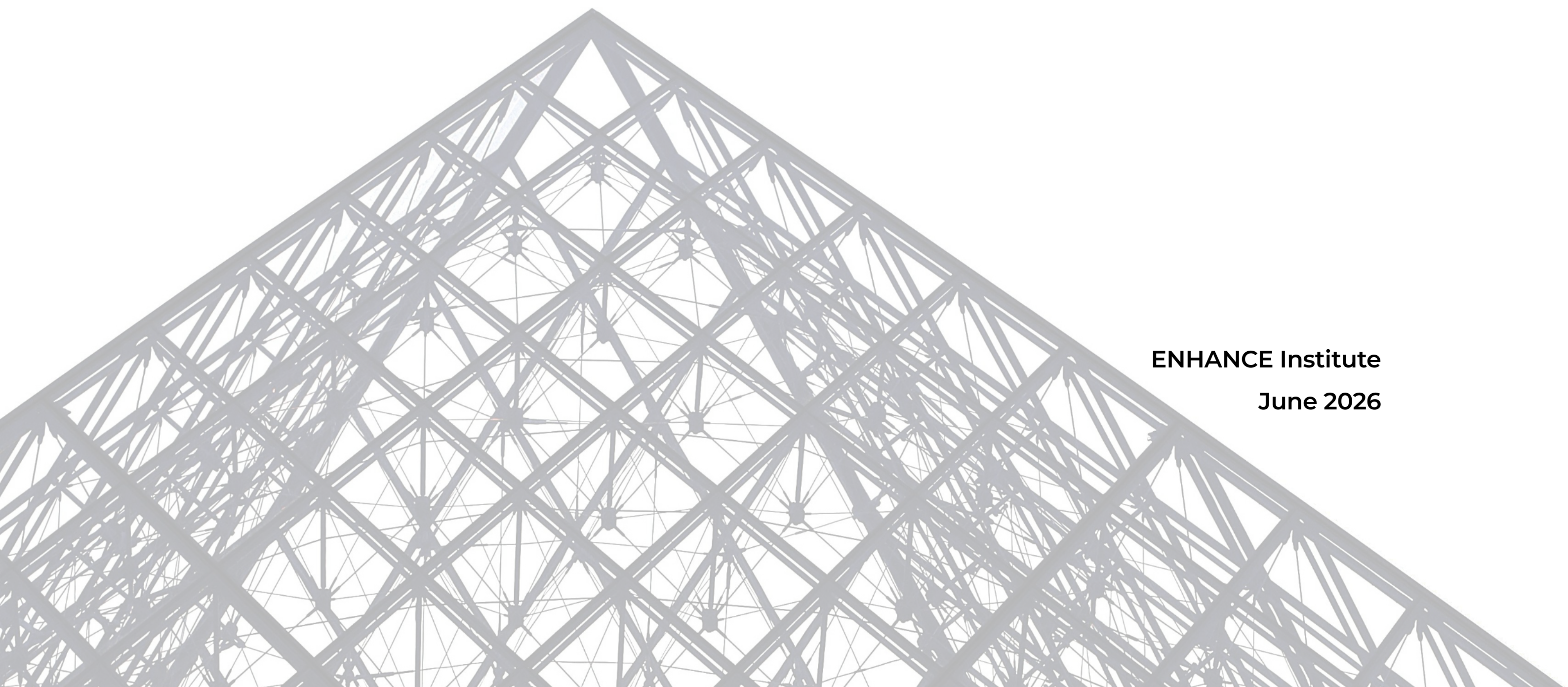
Output mode	**Public Output**	**Institutional Output**
Purpose	Provides a high-level screening result for public disclosure, initial review, or general communication.	Provides expanded diagnostic interpretation for clients, institutions, funders, buyers, or portfolio reviewers under controlled engagement.
Audience	Public readers, website visitors, general stakeholders, early-stage inquiries.	Climate finance institutions, project owners, buyers, portfolio managers, MDBs/DFIs, and approved institutional clients.
Shows	Project context, broad comparability status, screening result, public flag, and high-level interpretation.	Expanded interpretation, diagnostic reasoning, review priorities, and institution-facing screening notes.
Does not show	Internal parameters, thresholds, calibration values, diagnostic logic, or ENHANCE integration signals.	Core engine logic, protected calibration values, routing thresholds, proprietary equations, or restricted ENHANCE system logic.
Use case	Demonstrates whether a claim appears plausible, requires review, or materially diverges from the public screening reference.	Supports deeper due diligence, project review, claim revision, portfolio screening, or pre-certification preparation.
Output status	Public screening signal. Not a certified carbon credit.	Institutional diagnostic output. Not a certified carbon credit, legal determination, or investment recommendation.
Disclosure level	Public disclosure mode.	Controlled institutional disclosure mode.

The institutional output mode is not displayed in this public report. Its existence is disclosed to clarify that GeoCarb™ can support deeper institutional review, while protected parameters, routing thresholds, diagnostic rules, and model-integration logic remain non-disclosed.

GeoCarb™

Carbon Claim Screening for REDD+ and NbS

A Physical Plausibility and Comparability Screen for
NbS Claims Before Funding, Purchase, or Certification



ENHANCE Institute

June 2026

EXECUTIVE SUMMARY

Nature-based carbon claims often appear as a single number.

GeoCarb™ v2.0 first asks what that number means.

Before a carbon claim becomes capital — before it is used to justify funding, purchase, certification preparation, or portfolio value — it should pass a prior test: whether the claimed carbon outcome is physically plausible, correctly classified, and defensible within a declared screening boundary.

GeoCarb™ was developed by ENHANCE Institute as an ex-ante screening methodology for nature-based carbon claims. It does not issue carbon credits. It does not replace field MRV. It does not substitute for certification standards. Its role is earlier: to identify whether a design-stage carbon claim is coherent enough to become the basis for financial reliance.

The central problem is simple.

A carbon number may look precise because it is expressed in tCO_2e. But tCO_2e is a unit, not a carbon process.

The same unit may be used to describe newly created vegetation uptake, avoided emissions, preserved carbon stocks, forest conservation, restoration, or a mixed claim containing several processes at once. These claims do not have the same physical meaning. They do not share the same measurement boundaries, time horizons, reversal risks, or comparability conditions.

GeoCarb™ v2.0 addresses this problem through a classification-first screening architecture.

It does not compare first and explain later.

It first determines whether the claim can be directly screened within GeoCarb™'s core physical plausibility boundary, whether only part of the claim is directly screenable, or whether the claim requires a specialized diagnostic pathway before any screening-adjusted result can be produced.

This distinction is essential.

A growth-based restoration claim should not be treated as if it were an avoided-deforestation claim. A stock-preservation claim should not be treated as if it were newly created atmospheric removal. A newly planted or early-stage project should not be evaluated as if it had already reached mature or functionally established uptake potential.

GeoCarb™ v2.0 therefore applies different screening treatment depending on the carbon process behind the number.

For growth-based nature-based solution claims, GeoCarb™ screens whether the declared carbon outcome is physically plausible for the stated location, vegetation category, project scale, climate context, and current project stage.

For avoided-emission, conservation, or stock-preservation claims, GeoCarb™ does not force a direct comparison with vegetation uptake potential. It routes the claim through a protected diagnostic process designed to assess whether the structure behind the claim is coherent before it is treated as financial value.

For mixed claims, GeoCarb™ separates what can be screened within its declared boundary from what requires additional evidence, external MRV, or specialized treatment.

All outputs are reported as screening signals.

A **PASS** means the claim appears physically plausible within GeoCarb™'s declared screening boundary.

A **WARNING** means the claim may be plausible but requires additional evidence, review, or methodological clarification.

A **DISTORTION** means the claim materially diverges from the relevant screening reference or diagnostic result and should be reviewed before being used as a basis for funding, purchase, certification preparation, or portfolio value.

These outputs are not certification decisions.

A PASS does not issue a credit.
A WARNING does not invalidate a project.
A DISTORTION does not prove fraud or project failure.

GeoCarb™ provides a prior screening signal. It helps identify whether a carbon claim should proceed, be reviewed, be decomposed, or be revised before capital is committed.

The model has been benchmarked against satellite-derived vegetation productivity patterns and literature-supported ecological ranges. These benchmarking results support GeoCarb™'s use as a physical plausibility screening layer within its declared scope. They do not make GeoCarb™ a full ecosystem process model, a real-time satellite monitoring system, or a replacement for site-level MRV.

The public architecture of GeoCarb™ v2.0 is disclosed in this report. The protected engine is not.

Internal calibration parameters, routing thresholds, sensitivity ranges, risk modifiers, component-level calculation logic, and ENHANCE integration structures remain proprietary and are not disclosed. This boundary is intentional. GeoCarb™ is designed to be explainable without being replicable.

GeoCarb™ v2.0 therefore asks three questions before a carbon claim becomes capital:

1. **What kind of carbon process is being claimed?**
2. **Can that claim be screened within GeoCarb™'s declared boundary, or does it require specialized diagnostic treatment?**
3. **If screenable, is the claim physically plausible at this location, for this vegetation, at this growth stage, and at this scale?**

Only after those questions are answered should the claim be used to support funding, purchase, certification preparation, or portfolio value.

GeoCarb™ does not replace certification.
It protects claim quality before certification begins.

GeoCarb™ does not replace MRV.
It identifies where MRV should focus.

GeoCarb™ does not claim to measure everything.
It declares what it can screen, what it cannot screen, and when a carbon claim must be routed elsewhere before it can be trusted.

Classification before comparison.
Diagnosis before capital.

Section 1 — The Missing Question in NbS Carbon Finance

Climate finance appraisal asks many questions about a nature-based solution project.

Is the project design sound?
Does it have community buy-in?
Is the implementing entity credible?
Will it generate co-benefits?
Can it be registered under an approved standard?

Yet two questions are often asked too late.

The first is about meaning:

What kind of carbon process is this claim actually describing?

The second is about physical plausibility:

Is the claimed carbon outcome physically defensible at this location, for this vegetation, at this project stage, and at this scale?

These are not replacements for implementation review, institutional due diligence, community assessment, certification preparation, or MRV. Those questions remain necessary.

But they are not sufficient.

Before a carbon number is accepted as the basis for funding, purchase, certification preparation, or portfolio value, the claim behind that number must first be understood.

GeoCarb™ v2.0 asks that prior question.

The meaning problem

A tCO_2e figure can look precise.

But tCO_2e is a unit of measurement.
It is not, by itself, a description of a carbon process.

The same unit may be used to describe newly created vegetation uptake, avoided deforestation, preservation of existing carbon stocks, forest restoration, conservation, or a mixed claim containing several carbon processes at once.

These are not the same physical event.

They may differ in:

- what carbon process is being claimed;
- what carbon pools are included;
- whether the claim depends on a counterfactual baseline;
- whether the result is annual or cumulative;
- whether the claim depends on permanence;
- whether reversal risk is material; and
- whether the claim can be directly compared with a physical uptake estimate.

When these differences are not identified, the number may appear credible simply because the unit is familiar.

This is the first gap GeoCarb™ addresses.

It classifies the claim before screening it.

The physical problem

Carbon absorption in terrestrial ecosystems is not spatially uniform.

It depends on location, vegetation category, project area, climate conditions, project stage, and the physical limits of the ecosystem being claimed.

A newly planted agroforestry system should not be assessed as if it were already a mature or functionally established forest.

A claim derived from one ecological context should not be transferred to another site without testing whether the physical conditions are comparable.

A conserved forest stock should not be treated as if it were the same process as newly created atmospheric removal.

The physical gap is therefore not abstract.

Its consequences are financial and structural.

When public climate finance or private capital is committed to claims that have not been independently screened, scarce mitigation resources may be directed toward projects whose carbon outcomes are not achievable at the declared scale, or whose carbon process has been misunderstood before the claim becomes capital.

The GeoCarb™ response

GeoCarb™ v2.0 addresses both gaps through a classification-first screening architecture.

Before any physical comparison is made, GeoCarb™ first determines whether the claim can be screened directly within its declared physical boundary, whether only part of the claim can be screened directly, or whether the claim requires a specialized diagnostic pathway.

This protects the analysis from false comparability.

Growth-based claims, avoided-emission claims, stock-preservation claims, and mixed claims may all matter for climate mitigation. But they should not be treated as interchangeable simply because they are expressed in the same unit.

GeoCarb™ v2.0 does not compare first and explain later.

It asks:

What kind of carbon claim is this?
Can it be screened within GeoCarb™'s declared boundary?
If so, is it physically plausible at this location, for this vegetation, at this project stage, and at this scale?

Why this matters before capital is committed

If you are appraising an NbS funding proposal, you may be asked to trust a carbon number that has not yet been classified by carbon-process type or independently screened for physical plausibility.

If you are buying carbon credits from a nature-based project, the volume you are purchasing may depend on assumptions about uptake, avoidance, permanence, or stock preservation that deserve independent screening before being treated as financial value.

If you are designing an NbS project, your carbon claim may be physically overstated, dependent on a project stage not yet reached, or based on a carbon process that has not been clearly separated from other components of the claim.

In each case, the risk appears before implementation failure.

It appears when the carbon number first becomes credible enough to attract capital.

GeoCarb™ v2.0 is designed to intervene at that point.

It does not replace certification.
It does not replace field MRV.
It does not issue carbon credits.

It asks, before the claim becomes capital:

What kind of carbon process is being claimed — and is that claim physically defensible within a declared screening boundary?

Section 2 — Every Design-Stage Carbon Claim Is Already a Model

When a project developer writes that a project will sequester 50,000 tCO_2e per year, or that a programme will avoid 3.6 million tCO_2e per year, that number does not emerge from nowhere.

It is produced by assumptions.

Those assumptions may come from default factors, published studies, analogue projects, field estimates, national reference levels, remote-sensing interpretation, methodology guidance, or professional judgment.

In every case, a model has already been applied.

It may be explicit.
It may be embedded in a spreadsheet.
It may be inherited from a methodology.
It may be implied through a coefficient, benchmark, or reference scenario.

But the claim is never model-free.

The current system does not eliminate models from the design stage. It often relies on models that are difficult to inspect, difficult to compare, or insufficiently constrained by the physical conditions and carbon-process type of the actual project.

GeoCarb™ v2.0 does not argue that modelling should be removed.

It argues that the model behind a carbon claim should be made visible enough to be classified, screened, and tested before the claim becomes financial value.

The prior question

Before GeoCarb™ v2.0 tests whether a carbon number is physically plausible, it asks what the carbon number represents.

A declared claim of 50,000 tCO_2e per year from a newly planted agroforestry project is not the same kind of claim as a declared claim of 3.6 million tCO_2e per year from an avoided-deforestation programme.

The first may depend on future vegetation growth.
The second may depend on a counterfactual baseline.

They may share the same unit, but they do not describe the same carbon process.

They require different screening treatment.

If the same direct comparison is applied to both, the result may appear numerical but still be structurally misleading.

GeoCarb™ v2.0 resolves this before any number is compared.

What GeoCarb™ makes visible

GeoCarb™ does not disclose or replace the full internal model behind every carbon claim.

Instead, it asks whether the claim contains enough information to answer three public questions:

1. **What type of carbon process is being claimed?**
2. **Can that process be screened within GeoCarb™'s declared boundary?**
3. **Does the claimed outcome appear physically plausible for the declared project context?**

For growth-based claims, GeoCarb™ asks whether the declared uptake is defensible for the project's location, vegetation category, scale, climate context, and current stage of development.

For avoided-emission and conservation-type claims, GeoCarb™ asks whether the claim should be treated through a different diagnostic pathway before it is accepted as financial value.

For mixed claims, GeoCarb™ asks whether the claim has combined different carbon processes into one aggregate number.

This is the practical function of GeoCarb™ v2.0.

It does not make every calculation public.
It makes the claim's structure testable.

The practical difference

Without GeoCarb™	With GeoCarb™ v2.0
Carbon claim accepted as a single tCO_2e number	Carbon claim classified before screening
Carbon process may remain implicit	Carbon process is identified at the screening level
Different claim types may be treated as comparable	Different claim types are routed differently
Project stage may be overlooked	Current-stage defensibility is considered
Avoided emissions and removals may be merged in one figure	Different carbon meanings are separated before screening
Physical plausibility may remain untested at proposal stage	Claim is screened within a declared physical boundary
Output remains the declared number	Output becomes a screening signal with interpretation limits

Why this matters

A design-stage carbon claim is not just an environmental statement.

It can shape funding decisions.
It can support a purchase agreement.
It can influence portfolio value.
It can prepare a project for certification.
It can make a proposal appear more viable than it would otherwise appear.

That is why the claim must be screened before it becomes capital.

GeoCarb™ does not ask project developers, funders, or buyers to abandon existing standards.

It asks a prior question:

Has the number entering those standards been classified and physically screened before it becomes financial value?

That is the purpose of GeoCarb™ v2.0.

It does not certify the claim.
It does not replace MRV.
It does not issue credits.

It identifies whether the claim's underlying model is coherent enough, physically plausible enough, and comparable enough to be relied upon within a declared screening boundary.

Section 3 — What GeoCarb™ Is — and Is Not

Clarity of scope is essential.

GeoCarb™ v2.0 is designed to do two things well.

First, it identifies what kind of carbon claim is being made.

Second, it screens whether that claim is physically plausible, comparable, and defensible within a declared boundary.

It is not designed to do everything.

GeoCarb™ operates before certification, purchase, or capital commitment. Existing carbon standards, certification systems, registries, and MRV procedures play necessary roles in validation, monitoring, verification, registration, and credit issuance.

GeoCarb™ does not replace those systems.

It asks an earlier question:

Has the carbon claim entering those systems been correctly understood and physically screened before it becomes financial value?

GeoCarb™ does not replace certification.
It protects claim quality before certification, funding, or purchase begins.

What GeoCarb™ v2.0 is

1. A carbon-claim comparability screen

GeoCarb™ first determines whether a carbon claim can be meaningfully compared within its declared screening boundary.

Not every tCO_2e claim describes the same carbon process. A claim based on new vegetation growth, a claim based on avoided emissions, a claim based on forest conservation, and a mixed claim may require different treatment.

GeoCarb™ identifies this difference before screening the number.

2. A physical plausibility screening methodology

GeoCarb™ tests whether a nature-based carbon claim appears physically plausible for the declared project context.

That context may include location, vegetation category, project scale, project stage, climate conditions, and the type of carbon process being claimed.

The purpose is not to produce a certified credit volume.
The purpose is to determine whether the claim is physically defensible enough to proceed without review.

3. An ex-ante diagnostic layer

GeoCarb™ is designed for the proposal, appraisal, pre-purchase, and pre-certification preparation stage.

It operates before full implementation data are available and before a claim becomes the basis for financial reliance.

At this stage, the carbon number may already influence funding, purchase, portfolio valuation, or certification preparation. GeoCarb™ provides a prior screening signal before that influence becomes capital.

4. A pathway-specific screening system

GeoCarb™ does not apply one universal comparison to every carbon claim.

Growth-based claims are screened through a physical plausibility pathway.

Avoided-emission, conservation, stock-preservation, or mixed claims may require a different diagnostic pathway before any screening-adjusted result can be produced.

The specific internal routing rules, thresholds, and component-level calculations are not disclosed in this public report.

5. A screening support tool for institutions, buyers, and project designers

GeoCarb™ can support climate finance appraisal, carbon market due diligence, and project design quality assurance.

It helps users identify whether a claim appears plausible, requires further evidence, should be decomposed, or should be revised before it is relied upon.

What GeoCarb™ v2.0 is not

1. It is not a carbon-credit certification system

GeoCarb™ does not issue carbon credits.

It does not replace Verra, Gold Standard, CDM, GS4GG, national registries, or any other certification or crediting programme.

A GeoCarb™ PASS does not mean that a project will receive certification.

2. It is not a replacement for field MRV

GeoCarb™ does not replace field measurement, forest inventory, LiDAR survey, soil sampling, satellite monitoring, community-level verification, or third-party assessment.

It is a screening layer, not a final measurement system.

3. It is not a full ecosystem process model

GeoCarb™ does not fully simulate all ecological dynamics.

It does not claim to model every interaction among soil, nutrients, water, species competition, disturbance, pests, fire, or long-term ecosystem transition.

It screens carbon claims within a declared boundary.

4. It is not a real-time satellite monitoring system

GeoCarb™ uses calibrated screening logic and benchmarked physical relationships. It is not a live remote-sensing platform that measures current forest condition in real time.

Current project condition must still be confirmed through project-level MRV or independent evidence.

5. It is not a complete blue-carbon or peatland model

GeoCarb™ v2.0 may screen selected terrestrial or vegetation-related nature-based claims where they fall within its declared boundary.

However, blue-carbon sediment, peatland soil carbon, methane flux, water-table dynamics, and long-term burial carbon require specialized modules or external MRV evidence.

6. It is not a legal, regulatory, or investment decision tool

GeoCarb™ outputs are screening and diagnostic signals.

They should not be used as the sole basis for legal claims, investment decisions, contractual payment obligations, regulatory compliance, or carbon-credit issuance.

7. It is not a universal mitigation model

GeoCarb™ is not designed for solar, wind, energy efficiency, industrial process mitigation, CCS, methane capture, fuel switching, transport decarbonization, or other non-NbS mitigation pathways.

Those project types require different tools.

What changed from v1.1 to v2.0

GeoCarb™ v1.1 focused primarily on physical absorption plausibility.

GeoCarb™ v2.0 expands that role.

It now asks not only whether a claimed carbon outcome is physically plausible, but also whether the claim is the right kind of claim to be compared in the first place.

This is the core change.

GeoCarb™ v2.0 is no longer only a physical screen for vegetation uptake claims. It is a carbon-claim comparability and physical plausibility screen.

That means some claims may be screened directly.
Some may be screened only in part.
Some may require specialized diagnostic treatment before any screening-adjusted result can be produced.

This expansion does not mean GeoCarb™ measures everything.

It means GeoCarb™ is clearer about what can be screened, what cannot be screened directly, and when a claim must be routed elsewhere before it can be trusted.

Scope principle

GeoCarb™ v2.0 is strongest when used for what it was designed to do:

- identify the carbon process behind a claim;
- determine whether the claim is comparable within a declared boundary;
- screen physical plausibility before capital is committed;
- distinguish growth-based claims from avoided-emission or conservation-type claims;
- produce a screening-adjusted diagnostic signal; and
- clarify where MRV or specialized review should focus.

It is weakest when treated as a certification system, a complete ecosystem simulator, or a substitute for field MRV.

That boundary is not a weakness.

It is the condition that makes the screening result credible.

Section 4 — Why Design-Stage Estimates Are Structurally Limited

Design-stage carbon estimates are necessary.

Without them, nature-based solution projects cannot be appraised, financed, compared, or prepared for certification.

The problem is not that design-stage estimates exist.
The problem is that they are often treated as if they are already physically constrained, taxonomically precise, and comparable across project types.

They are not.

Design-stage carbon claims are built before full implementation evidence exists. They often rely on default values, analogue projects, expected growth rates, reference scenarios, or professional assumptions. These inputs may be useful. But they do not automatically prove that the declared carbon outcome is physically plausible at the project site, for the project stage, or for the type of carbon process being claimed.

GeoCarb™ v2.0 addresses this limitation by imposing two prior constraints.

First, it asks whether the carbon process has been correctly understood.

Second, it asks whether the claim is physically defensible within a declared screening boundary.

The taxonomic limitation

A design-stage carbon claim is often expressed as one number.

That number may look clean.
But the process behind it may not be.

A single tCO_2e claim may refer to new vegetation growth, avoided emissions, conservation of existing stocks, restoration of degraded land, or a mixture of more than one carbon process.

If those meanings are not separated at the screening stage, the claim may appear more comparable than it really is.

This is a taxonomic limitation.

A growth-based claim should not be screened in the same way as an avoided-emission claim. A claim based on conserving an existing carbon stock should not be treated as if it were the same as newly created atmospheric removal. A mixed claim should not be accepted as one undifferentiated number if its components depend on different physical assumptions.

GeoCarb™ v2.0 does not assume that all tCO_2e claims are interchangeable.

It first determines whether the claim can be screened directly, only in part, or only through a separate diagnostic pathway.

This prevents false comparability before physical screening begins.

The physical limitation

Even when the carbon process is correctly understood, the claim may still be physically unconstrained.

Three patterns are common.

First, default factors provide broad reference values.

Default factors and methodology-level reference values are useful. They provide recognized categories, benchmark ranges, and accounting structures.

But they are not, by themselves, a site-specific physical test.

A broad vegetation category may not reflect the project's location, climate conditions, project area, vegetation composition, or development stage. A value that is reasonable in one ecological context may become overstated when transferred to another.

GeoCarb™ v2.0 does not reject default values.

It asks whether the value used in a project claim remains physically plausible in the declared project context.

Second, analogue projects can import hidden assumptions.

Design-stage estimates often rely on comparable projects, regional examples, or observed sequestration rates from other contexts.

This is understandable. At the design stage, project-specific evidence may be limited.

But analogues can mislead when the basis of comparison is not tested.

A high-performing project in one watershed, one climate regime, or one development stage may not provide a defensible reference for another site. The analogue may be real, but the transfer may not be physically appropriate.

GeoCarb™ v2.0 therefore asks whether the comparison itself is defensible before the claim is accepted.

Third, upper-bound assumptions can become financial assumptions.

Design-stage claims are often made in contexts where higher projected carbon outcomes can strengthen the apparent viability of a proposal.

This does not require bad intent.

It can occur because there is no independent screening layer forcing the claim back into the physical limits of the site.

Without such a constraint, theoretical potential can be mistaken for current-stage performance. Broad biome averages can be applied to local projects. Expected future outcomes can be treated as if they were already defensible at the design stage. Avoided-emission claims can be presented in a way that makes them appear equivalent to additional removals.

The result is a structural risk:

capital may be allocated on the basis of a carbon number that has not yet been physically disciplined.

Project stage as a hidden constraint

For afforestation, reforestation, restoration, agroforestry, and related projects, project stage is one of the most important omitted dimensions.

A newly planted system cannot be assessed as if it were already mature or functionally established.

An early-stage project may eventually develop significant uptake potential. But that future potential is not the same as current-stage defensibility.

If a project is still in the early years of development, the screening basis must reflect that fact.

GeoCarb™ v2.0 therefore distinguishes between long-run physical potential and current-stage plausibility. It does not assume that a project has already reached the performance level it may only achieve later.

This is one of the most important safeguards in the model.

Counterfactual claims as a separate limitation

Avoided-emission and conservation-type claims introduce a different limitation.

They do not primarily ask how much new vegetation uptake is possible.

They ask what would have happened without the project.

That question depends on a counterfactual scenario. It may involve assumptions about future forest loss, degradation, land-use change, biomass fate, permanence, or continued stock protection.

Those assumptions may be reasonable.
But they should not be treated as equivalent to new atmospheric removal.

GeoCarb™ v2.0 therefore does not force counterfactual claims into the same screening pathway as growth-based uptake claims.

It first identifies whether the claim requires separate diagnostic treatment before a screening-adjusted result can be produced.

This protects the analysis from asking the wrong physical question.

GeoCarb™ imposes both constraints

GeoCarb™ v2.0 imposes two constraints before a design-stage carbon claim is accepted as financially meaningful.

The first is a meaning constraint:

What kind of carbon process is this claim describing?

The second is a physical constraint:

Is that claim plausible in the declared project context and within GeoCarb™'s screening boundary?

This is the difference between accepting a carbon number and screening the claim behind it.

GeoCarb™ does not ask whether a project developer believes the number is achievable.

It asks whether the declared claim can survive classification, comparability, and physical plausibility screening before it becomes the basis for funding, purchase, certification preparation, or portfolio value.

That is the structural limit GeoCarb™ v2.0 is designed to impose.

Section 5 — Model Architecture: Classification Before Comparison

GeoCarb™ v2.0 is built on a classification-first architecture.

The purpose of this architecture is simple: a carbon number should not be accepted, compared, or financially relied upon until the model first understands what kind of carbon claim that number represents.

A nature-based project may claim removals, avoided emissions, preserved carbon stocks, future sink capacity, or a combination of these. Each may be expressed in tCO_2e. But the same unit does not mean the same carbon process.

GeoCarb™ v2.0 therefore does not begin by asking whether a number is high or low.

It begins by asking:

What is this number actually claiming?

Only after that question is answered does GeoCarb™ select the appropriate screening pathway.

A public architecture, not a disclosed engine

The public logic of GeoCarb™ v2.0 can be summarized in three steps.

First, the claim is classified.
Second, the claim is routed to the appropriate screening pathway.
Third, the result is reported as a screening signal within a declared boundary.

This public architecture allows readers to understand why the model is logically disciplined. It does not disclose the protected engine that determines internal thresholds, calibration values, routing conditions, parameter weights, or component-level calculations.

In other words, this report explains **how GeoCarb™ thinks**, not **how GeoCarb™ is reconstructed**.

Step 1 — Classify the claim

GeoCarb™ first identifies the nature of the carbon claim.

A growth-based restoration claim is not the same as an avoided-deforestation claim. A claim based on preserving an existing forest stock is not the same as a claim based on newly planted vegetation. A mixed claim may contain more than one carbon process inside a single tCO_2e figure.

If those differences are ignored, the analysis may appear numerical but remain physically misleading.

GeoCarb™ v2.0 avoids this by classifying the claim before comparison. The classification determines whether the claim can be screened directly, partially, or only through a specialized diagnostic pathway.

This is the first protection against false comparability.

Step 2 — Route the claim

After classification, GeoCarb™ routes the claim.

Claims that describe growth-based uptake can be screened against GeoCarb™'s core physical plausibility layer. This pathway evaluates whether the declared carbon outcome is plausible for the stated location, vegetation category, project scale, climate context, and project stage.

Claims that describe avoided emissions, stock preservation, or forest conservation are treated differently. They are not forced into a direct comparison with vegetation uptake potential. Instead, they are routed to a diagnostic pathway designed to examine the structure behind the claim before a screening-adjusted result is produced.

Mixed claims may require separation before screening. Where a single number contains both comparable and non-comparable elements, GeoCarb™ does not treat the number as one undifferentiated result. It identifies the parts that can be screened within the model boundary and flags the parts that require specialized evidence, external MRV, or additional review.

The routing process is central to GeoCarb™ v2.0.

It prevents a common error: treating avoided emissions, stock preservation, and additional removals as if they were interchangeable simply because they share the same unit.

Step 3 — Screen within a declared boundary

Once routed, the claim is screened within the appropriate boundary.

For growth-based claims, GeoCarb™ applies a physical plausibility screen. The model considers whether the claimed uptake is defensible for the location, vegetation category, project scale, climate context, and current growth stage.

This is important because a newly planted or early-stage project should not be evaluated as if it had already reached mature-system uptake potential.

For avoided-emission and conservation-type claims, GeoCarb™ applies a different diagnostic logic. The relevant question is not simply how much new vegetation uptake is possible. The question is whether the avoided-emission claim is structurally defensible, internally consistent, and supported by a plausible counterfactual baseline.

These pathways are different because the carbon processes are different.

GeoCarb™ v2.0 does not collapse them into one formula.

Screening outputs

GeoCarb™ reports results as screening signals.

Output	Public meaning
PASS	The claim appears physically plausible within GeoCarb™'s declared screening boundary.
WARNING	The claim may be plausible but requires additional review, evidence, or methodological clarification.
DISTORTION	The claim materially diverges from the screening reference or diagnostic result and should be reviewed before funding, purchase, or certification preparation.

These outputs are not certification decisions.

A PASS does not issue a credit.
A WARNING does not invalidate a project.
A DISTORTION does not prove fraud.

The output indicates whether the claim should proceed, be reviewed, be decomposed, or be revised before it becomes financial value.

What is deliberately not disclosed

GeoCarb™ is designed to be explainable without being replicable.

For that reason, this report does not disclose:

- internal calibration parameters;
- vegetation-specific coefficients;
- routing thresholds;
- sensitivity ranges;
- protected weighting structures;
- risk-side adjustment values;
- component-level equations;
- proprietary decision rules;
- code implementation; or
- ENHANCE internal integration logic.

Those elements remain restricted.

The public report explains the model's screening philosophy, claim-classification discipline, and output interpretation. It does not provide the information required to reproduce the engine.

Why this structure matters

GeoCarb™ v2.0 creates trust by making its boundaries explicit.

It tells the reader what the model does.
It tells the reader what the model does not do.
It explains why different carbon claims require different screening pathways.
It clarifies that screening-adjusted results are not certified credits.
It separates claim plausibility from credit issuance.

That is the level of transparency needed for institutional trust.

But the protected model remains non-disclosed.

GeoCarb™ therefore follows a public rule:

The logic may be explained.
The engine is not disclosed.

Architectural principle

GeoCarb™ v2.0 does not compare first and explain later.

It classifies first.
It routes second.

It screens third.
It reports within a declared boundary.

This is the public architecture.

The proprietary architecture — the calibration logic, thresholds, internal parameters, sensitivity bands, and component-level computation rules — remains protected by ENHANCE Institute.

That distinction is intentional.

It allows GeoCarb™ to be trusted by readers, used by institutions, and defended methodologically, without making the system replicable by external actors.

Section 6 — Benchmarking and Boundary Testing

GeoCarb™ v2.0 is designed as an ex-ante screening methodology.

It is not a post-implementation certification system.
It does not replace field MRV, third-party validation, registry approval, ecological survey, or credit issuance.

For that reason, this section does not present GeoCarb™ as a certification-grade measurement tool. Instead, it presents three forms of methodological support:

1. **model-level benchmarking** against independent satellite-derived vegetation productivity patterns;
2. **field case signals** showing how GeoCarb™ behaves when applied to real project contexts; and
3. **boundary testing** showing where GeoCarb™ should not be applied without additional evidence or specialized treatment.

This distinction matters.

A credible screening model should not only show where it performs well.
It should also declare where its results must be limited, qualified, or routed elsewhere.

6.1 Model-Level Benchmarking

GeoCarb™ Core was benchmarked against satellite-derived net primary productivity patterns across multiple latitude bands and supported vegetation categories.

The purpose of this benchmarking was to test whether GeoCarb™ Core captures the broad physical relationship between vegetation productivity and geographic-climatic structure.

It was not intended to disclose internal calibration parameters or reproduce the protected model engine.

Benchmarking scope	Result
Aggregated vegetation categories	$R^2 = 0.91$
Wildflower-specific benchmark	$R^2 = 0.9883$
Wildflower RMSE	34.0 gC/m²/yr
Latitude bands tested	27

The aggregated $R^2 = 0.91$ result indicates that GeoCarb™ Core captures a substantial share of observed variation in satellite-derived vegetation productivity patterns across the tested latitude structure and vegetation categories.

The wildflower-specific benchmark provides a stronger fit within one vegetation category, with $R^2 = 0.9883$ and RMSE of 34.0 $gC/m^2/yr$.

These results support the use of GeoCarb™ Core as a physical plausibility screening layer within its declared boundary.

They do not mean that GeoCarb™ is a full ecosystem process model.
They do not mean that GeoCarb™ replaces field measurement.
They do not mean that GeoCarb™ produces certified credit volumes.

The conclusion is narrower and more defensible:

GeoCarb™ Core provides a benchmark-supported physical screening reference for vegetation uptake claims within its declared model boundary.

6.2 Field Case Signal A — TIST Kenya

Comparable growth-based pathway

As a field-level consistency signal, GeoCarb™ was applied to the TIST Kenya agroforestry programme — an independent programme with no connection to ENHANCE.

TIST Kenya is a farmer-led agroforestry programme established in 1999, involving smallholder farmers planting and maintaining trees on degraded farmland across the Kenya Central Highlands. The programme has been independently verified under the Verified Carbon Standard and assessed by researchers at the University of Exeter.[1]

This case is useful because it allows GeoCarb™ Core to be compared with an external long-run programme-level sequestration proxy.

Input	Value	Source
Latitude	−0.5°	Kenya Central Highlands
Vegetation category	Agroforestry	TIST smallholder programme
Project area	18,278 ha	Exeter University assessment
Growth-stage reference	Early-stage classification	Conservative boundary rule

Metric	Value
GeoCarb™ Core potential	62,277.7 tCO_2e/yr
Growth-stage-adjusted reference	17,437.8 tCO_2e/yr
Annualized comparison proxy	~60,000 tCO_2e/yr[2]
Core deviation against proxy	3.8%
Core benchmark signal	Consistent

The GeoCarb™ Core estimate of 62,277.7 tCO_2e/yr differs by +3.8% from the annualized comparison proxy of approximately 60,000 tCO_2e/yr.[2]

This provides a field-level consistency signal supporting the GeoCarb™ Core estimate as a long-run or programme-level physical uptake reference.

However, this does not mean that the current-stage screening reference is equivalent to the long-run programme proxy.
The growth-stage-adjusted reference of 17,437.8 tCO_2e/yr reflects the project's current-stage classification under GeoCarb™ v2.0. It is the relevant comparison basis when a design-stage claim is being evaluated at the current project stage.

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[1] Buxton, J., Lomax, G., Powell, T., et al. "Carbon sequestration and regreening by Kenyan smallholder farmers." University of Exeter, Global Systems Institute. TIST Kenya programme assessment: 18,278 hectares confirmed showing positive greening trend; cumulative sequestration exceeding 1.2 million tonnes CO_2 since 2008.

[2] The annualized comparison proxy of ~60,000 tCO_2e/yr is derived from the Exeter University independent assessment that TIST Kenya farmers sequestered over 1.2 million tonnes of CO_2 in trees planted since 2008, divided across the programme period. This figure is used as a proxy for illustrative comparison purposes only and is not a formally reported annual sequestration figure.

This distinction is central to GeoCarb™ v2.0.

GeoCarb™ Core supports a long-run physical uptake reference.
The growth-stage-adjusted reference supports current-stage defensibility.

The TIST Kenya case therefore supports two conclusions:

1. GeoCarb™ Core is directionally consistent with an independent programme-level sequestration proxy; and
2. GeoCarb™ v2.0 prevents early-stage or partially established projects from being assessed as if they had already reached full long-run uptake potential.

6.3 Field Case Signal B — Laos REDD+

Non-directly-comparable avoided-emission pathway

As a field-level signal for the non-directly-comparable pathway, GeoCarb™ v2.0 was applied to a large-scale avoided-deforestation context in Laos.

This case tests the routing logic for avoided-emission claims.

Because the declared claim is an avoided-emission claim, GeoCarb™ Core is not used as a direct comparator. A REDD+ or avoided-deforestation claim does not primarily ask how much new vegetation uptake is physically possible. It asks what emissions or stock losses were avoided relative to a counterfactual baseline.

GeoCarb™ v2.0 therefore routes the claim through a protected avoided-emission diagnostic pathway.

Input	Value	Source
Latitude	+18.0°	Northern Laos
Vegetation category	Tropical forest	Project documentation
Project area	13,500,000 ha	Project documentation
Project type	REDD+ / avoided deforestation	Project screening input
Declared claim	3,669,638 tCO_2e/yr	Project screening input

Diagnostic result	Value
Screening-adjusted total	1,478,405 tCO_2e/yr
Declared claim	3,669,638 tCO_2e/yr
Deviation	-59.7%
Final screening signal	DISTORTION[3]

The protected diagnostic pathway produced a screening-adjusted result of 1,478,405 tCO_2e/yr against a declared claim of 3,669,638 tCO_2e/yr.

The −59.7% deviation triggered a DISTORTION signal.[3]

This does not mean that the project is fraudulent.
It does not mean that the project has no mitigation value.

[3] The Laos REDD+ illustrative screening case applies GeoCarb™ v2.0's protected avoided-emission diagnostic pathway to a publicly documented avoided-deforestation project context. The screening-adjusted result of 1,478,405 tCO_2e/yr is a GeoCarb™ internal screening output. It does not constitute a certified, independently verified, or registry-approved carbon volume.

It means that, under GeoCarb™ v2.0's conservative screening architecture, the declared avoided-emission claim materially diverges from the defensible screening-adjusted result.

The Laos case demonstrates why GeoCarb™ v2.0 does not directly compare avoided-emission claims with Core vegetation uptake potential.

A direct comparison would ask the wrong question.

The appropriate screening question is:

Which parts of the avoided-emission claim remain defensible after the claim is routed through the appropriate diagnostic pathway?

6.4 Boundary Cases — Scope Definition, Not Model Failure

Three additional cases were assessed as boundary cases.

They are included not as model failures, but as scope-definition signals.

A model that cannot state what it does not measure is less trustworthy than one that can.

Case	Test result	Reason for boundary classification
Mikoko Pamoja, Kenya, mangrove, −4.4°	Large negative divergence	Blue-carbon sediment and soil carbon dominate the claim. These pools remain outside GeoCarb™ v2.0 Core scope by design.
ReforesTerra, Brazil, tropical forest, −15°	Prior test limited by v1.1 structure	The earlier test was conducted under the v1.1 screening structure. Retesting under v2.0 or a higher-resolution biophysical module is recommended before drawing a final conclusion.
Tond Tenga, Burkina Faso, agroforestry, 12.5°N	Material divergence	Registered in 2025; insufficient verified performance history for use as a field comparison benchmark.

These cases clarify the boundary between three different conditions:

1. **Model failure** — the model should have worked but did not;
2. **Scope boundary** — the claim falls outside the model's declared screening domain; and
3. **Data maturity limitation** — the project does not yet have enough verified performance history for benchmarking.

GeoCarb™ v2.0 treats these differently.

Blue-carbon sediment claims and peatland soil-carbon systems require dedicated modules.
Recently registered projects may require more time, MRV, or verified performance data.
Projects previously tested under v1.1 may require retesting under v2.0 before conclusions are finalized.

Boundary identification is not a weakness.

It is part of methodological credibility.

6.5 Methodological Boundary Statement

GeoCarb™ v2.0 is designed for:

- design-stage screening of terrestrial nature-based carbon claims;
- growth-based removal and restoration claims within its declared screening boundary;
- avoided-emission and conservation-type claim diagnostics where direct comparison would be misleading;
- current-stage defensibility assessment using project-stage adjustment; and
- country-context downside-risk and reversal-risk screening within ENHANCE's protected risk architecture.

GeoCarb™ v2.0 does not model:

- full soil organic carbon dynamics;
- blue-carbon sediment storage;
- peatland methane flux;
- real-time satellite monitoring;
- post-implementation MRV;
- legal entitlement to carbon credits; or
- certification-grade credit issuance.

These are declared boundaries, not hidden assumptions.

GeoCarb™ v2.0 is strongest when used as intended: an ex-ante physical plausibility and comparability screen before a carbon claim becomes the basis for funding, purchase, certification preparation, or portfolio value.

Section 7 — Relationship to Certification Standards and MRV

One of the most important questions about GeoCarb™ is whether it conflicts with established carbon standards, certification pathways, or MRV systems.

It does not.

GeoCarb™ v2.0 is not designed to replace certification standards, carbon-crediting programmes, national registries, project-level MRV, third-party validation, or verification procedures.

Those systems play a necessary role. They define methodologies, validation requirements, monitoring plans, reporting procedures, verification standards, and issuance conditions within established carbon-crediting frameworks.

GeoCarb™ operates in a different temporal position.

It asks an earlier question:

Before a claim enters funding, purchase, portfolio valuation, or certification preparation, is the carbon number physically plausible, correctly classified, and structurally defensible?

This is not a substitute for certification.

It is a prior screening function.

GeoCarb™ operates in a different position.

It asks an earlier question:

Before a claim enters funding, purchase, portfolio valuation, or certification preparation, is the carbon number physically plausible, taxonomically coherent, and structurally defensible?

This is not a substitute for certification.

It is a prior screening function.

The temporal position

GeoCarb™ v2.0 operates before a carbon claim becomes financial value.

It is designed for the proposal, appraisal, pre-purchase, or pre-certification preparation stage — when the project's carbon claim is already influential, but before full implementation evidence, field MRV, or credit issuance exists.

The position can be understood as follows:

Project concept / funding proposal
↓
Design-stage carbon claim
↓
GeoCarb™ v2.0 screening

(classification + pathway-specific plausibility test)
↓
Funding / purchase / portfolio / preparation decision
↓
Registration or certification preparation
↓
Implementation and monitoring
↓
Validation / verification under applicable standards
↓
Credit issuance, where applicable

GeoCarb™ is not positioned above certification standards.

It is positioned before the point at which an untested or misclassified carbon claim becomes the basis for capital allocation.

Its purpose is to help ensure that the carbon number entering the next stage is physically defensible and correctly understood.

The complementary relationship

Certification standards and GeoCarb™ ask different questions.

Certification standards ask, within their approved methodologies and procedures:

Does the project meet the requirements of the applicable standard, and can the monitored outcome be validated, verified, and issued under that framework?

GeoCarb™ asks earlier:

Before the claim reaches that stage, is the claimed carbon outcome physically plausible and correctly classified at the design stage?

These are complementary functions.

A GeoCarb™ PASS does not certify a project.
A GeoCarb™ WARNING does not invalidate a project.
A GeoCarb™ DISTORTION does not prove misconduct.
A certified project is not automatically exempt from design-stage plausibility scrutiny.

GeoCarb™ provides a screening signal.
Certification provides a formal crediting pathway.

They are not substitutes.

The relationship to MRV

GeoCarb™ v2.0 does not replace MRV.

It does not replace:

- field measurement;

- forest inventory;
- LiDAR assessment;
- satellite monitoring;
- soil sampling;
- community-level verification;
- third-party validation; or
- project-level performance monitoring.

MRV is necessary because real-world project performance must be measured.

GeoCarb™ operates before full MRV evidence exists.

At the design stage, the project may already be making a carbon claim, but the field evidence required to confirm that claim may not yet be available. GeoCarb™ provides an independent screening reference that helps identify whether the claim is plausible enough to proceed, whether it requires methodological review, or whether it should be decomposed before being treated as financial value.

GeoCarb™ does not tell MRV what it must find.

It helps identify what MRV should pay attention to.

For example:

- if a growth-based claim appears high relative to the project's current stage, MRV should focus on growth-stage evidence and survival rates;
- if a REDD+ claim depends heavily on avoided stock loss, MRV should scrutinize baseline assumptions and forest-loss evidence;
- if a claim depends on biomass fate, MRV should examine whether harvested biomass would likely be burned, decomposed, stored, or converted into long-lived products;
- if a claim includes future sink capacity, MRV should test whether that sink function is actually supported by the forest condition and permanence structure.

In this sense, GeoCarb™ does not replace MRV.

It helps structure better MRV questions before capital is committed.

Why this distinction matters

Many carbon claims become financially meaningful before they are fully measured.

A project proposal can attract funding.
A pre-purchase agreement can be signed.
A portfolio value can be estimated.
A certification pathway can be prepared.
A project can be promoted on the basis of projected tCO_2e.

At that stage, the carbon number has not yet become an issued credit.
But it may already have become capital.

GeoCarb™ v2.0 intervenes at that point.

It asks whether the claim should be trusted enough to proceed.

If the claim is physically plausible, GeoCarb™ can support confidence before the formal certification or MRV process continues.

If the claim is uncertain, GeoCarb™ can identify where additional evidence is needed.

If the claim is materially divergent, GeoCarb™ can signal that review is required before the number is used as a basis for funding, purchase, certification preparation, or portfolio valuation.

One-line summary

GeoCarb™ before certification — not instead of certification.

GeoCarb™ before full MRV — not instead of MRV.

GeoCarb™ before capital — not after the claim has already shaped the decision.

Section 8 — Application Domains

GeoCarb™ v2.0 is applicable wherever a nature-based carbon claim is assessed before it becomes financial value.

A carbon claim may support a funding proposal, a credit purchase, a project valuation, a portfolio allocation, or a certification preparation process. In each case, the same prior question applies:

Has the carbon claim been classified, screened, and physically bounded before capital is committed to it?

GeoCarb™ v2.0 is designed to answer that question across four primary application domains.

8.1 Climate Finance Pre-Screening

Climate finance institutions, multilateral development banks, bilateral agencies, accredited entities, and public funding bodies may use GeoCarb™ v2.0 to screen nature-based carbon claims before approval or investment decisions are made.

At the proposal stage, carbon claims are often influential even though full implementation data do not yet exist. A projected tCO_2e figure may shape the perceived mitigation value of a project, the justification for concessional finance, or the ranking of competing proposals.

GeoCarb™ provides a prior screening layer.

For growth-based projects such as afforestation, reforestation, restoration, agroforestry, and assisted natural regeneration, GeoCarb™ assesses whether the declared claim is physically plausible for the location, vegetation category, project scale, and current growth stage.

For avoided-emission, conservation, or REDD+ type claims, GeoCarb™ does not force a direct comparison with vegetation uptake potential. Instead, the claim is routed through a protected diagnostic pathway that examines whether the claim structure is coherent before a screening-adjusted result is produced.

This allows climate finance decision-makers to identify claims that appear defensible, claims that require further review, and claims that should not be relied upon without methodological clarification.

GeoCarb™ does not approve or reject projects.
It screens whether the carbon number entering the appraisal process is physically and structurally defensible.

8.2 Carbon Market Due Diligence

Voluntary carbon market participants, credit buyers, rating agencies, portfolio managers, and project developers may use GeoCarb™ v2.0 to assess whether a nature-based carbon claim is physically plausible before purchase, portfolio inclusion, or certification preparation.

This is especially important because carbon credits and projected carbon outcomes are often expressed in the same unit, even when they arise from different carbon processes.

A tonne associated with avoided deforestation is not physically identical to a tonne associated with new vegetation growth. A tonne linked to stock preservation is not the same process as a tonne linked to additional atmospheric removal. These distinctions affect permanence, reversal risk, measurement boundaries, and financial defensibility.

GeoCarb™ v2.0 makes these distinctions explicit at the due diligence stage.

Its output is not a credit rating, certification decision, or legal conclusion. It is a screening signal that helps buyers and reviewers ask better questions before treating a carbon claim as financial value.

8.3 Project Design Quality Assurance

Project developers, consultants, and design teams may apply GeoCarb™ v2.0 during project preparation.

This use case is preventive.

Rather than waiting for external reviewers, validators, buyers, or funders to challenge a carbon estimate, a project team can screen its own claim before submission.

This is particularly relevant where:

- growth-stage assumptions may not have been clearly adjusted;
- mature-system performance may have been applied to an early-stage project;
- avoided-emission and removal components may have been aggregated into a single number;
- carbon pool boundaries may not have been clearly declared;
- REDD+ or conservation claims may depend on a counterfactual baseline requiring additional scrutiny;
- project-level MRV has not yet been designed around the most sensitive assumptions.

A GeoCarb™ PASS can support confidence that the claim is physically plausible within the declared screening boundary.

A WARNING can help the design team identify what additional evidence or clarification is needed.

A DISTORTION signal can indicate that the claim should be revised, decomposed, or re-supported before it is submitted externally.

In this role, GeoCarb™ is not a barrier to project design.
It is a design-quality safeguard.

8.4 Portfolio and Geographic Screening

Governments, investors, climate finance allocators, development institutions, and portfolio managers may use GeoCarb™ v2.0 to assess the plausibility profile of a pipeline of nature-based carbon claims.

This application is useful when multiple projects are being compared across regions, vegetation categories, project types, and carbon-claim structures.

GeoCarb™ does not rank projects only by declared tCO_2e volume.
It asks whether those volumes are physically plausible, correctly classified, and comparable within a declared boundary.

This matters because a portfolio may appear strong if judged only by headline carbon volume, while containing claims that differ substantially in process type, growth-stage maturity, permanence exposure, and screening defensibility.

GeoCarb™ supports more disciplined portfolio review by distinguishing:

- growth-based removal claims from avoided-emission claims;
- current-stage uptake from long-run potential;
- comparable claims from partially comparable or non-directly comparable claims;
- physically plausible claims from claims requiring further review.

Country-context risk adjustment may also be applied within ENHANCE's protected screening architecture. Public outputs disclose the effect as a risk-context screening signal, not the internal risk logic or calibration structure.

8.5 What GeoCarb™ Enables

Across these domains, GeoCarb™ enables a more disciplined sequence.

Instead of asking only:

How many tCO_2e does the project claim?

GeoCarb™ asks:

1. **What kind of carbon process is being claimed?**
2. **Is the claim comparable within GeoCarb™'s declared screening boundary?**
3. **If comparable, is it physically plausible at this location, for this vegetation, at this growth stage, and at this scale?**
4. **If not directly comparable, has the claim been routed through the appropriate diagnostic pathway before being treated as financial value?**

This sequence matters because all design-stage carbon claims are already models. The question is not whether a model is being used. It always is.

The question is whether that model has been made explicit enough to be classified, screened, and physically constrained before it becomes the basis for capital allocation.

8.6 Closing Statement

Nature-based carbon claims are not spatially, physically, or taxonomically uniform.

They are structured by location, vegetation class, climate conditions, growth stage, carbon-process type, carbon-pool boundary, permanence, and reversal risk.

These factors can be screened before capital is committed.

The structural gap that GeoCarb™ v2.0 addresses is not a problem of bad intentions. It is a problem of missing infrastructure: the absence of an independent mechanism that classifies what a carbon claim means before testing whether it is physically defensible.

GeoCarb™ v2.0 provides that screening infrastructure.

It does not replace certification.
It does not replace MRV.
It does not issue carbon credits.

It asks whether the claim is ready to become capital.

Structure before execution.
Classification before comparison.
Diagnosis before capital.

ENHANCE operates a structured GeoCarb™ screening pathway for NbS funding proposals, carbon-credit portfolios, and project pipeline assessments.

Inquiries may be directed to:

contact@enhance-institute.org

How to Use GeoCarb™ v2.0

Who is GeoCarb™ for?

GeoCarb™ v2.0 is designed for institutions and professionals that evaluate, fund, purchase, or design nature-based carbon claims before capital is committed.

GeoCarb™ does not replace certification, MRV, or project-level due diligence. It provides an earlier screening layer: a way to assess whether a declared carbon claim is physically plausible, correctly classified, and defensible within a declared screening boundary before it becomes the basis for funding, purchase, certification preparation, or portfolio value.

01. Climate Finance Institutions

Who:
Climate funds, MDBs, DFIs, bilateral development agencies, accredited entities, and public finance institutions.

Use case:
Pre-screening nature-based carbon claims in funding proposals before approval or investment decisions.

Purpose:
To assess whether the carbon claim in a proposal appears physically plausible and structurally coherent before it enters deeper appraisal, negotiation, or financing stages.

What GeoCarb™ provides:

- carbon-claim comparability screening;
- physical plausibility assessment for growth-based NbS claims;
- avoided-emission diagnostic screening for REDD+ and conservation-type claims;
- high-level screening flag: PASS / WARNING / DISTORTION;
- guidance on where further MRV, evidence, or due diligence should focus.

02. Carbon Market Participants

Who:
Credit buyers, portfolio managers, rating agencies, brokers, project reviewers, and carbon market due diligence teams.

Use case:
Due diligence on nature-based carbon claims before purchase, pricing, portfolio inclusion, or certification preparation.

Purpose:
To understand whether the carbon process behind a claim is physically defensible and correctly classified before the claim is treated as financial value.

What GeoCarb™ provides:

- claim-type and comparability screening;
- physical plausibility signal for declared NbS carbon outcomes;
- distinction between growth-based removal, avoided-emission, conservation, and mixed claims;
- screening flag as a pre-purchase integrity signal;
- indication of whether further review or independent evidence is required.

03. Project Developers

Who:
NbS project designers, carbon project consultants, implementing entities, and proposal development teams.

Use case:
Design-stage quality assurance before submission for funding, purchase, registration, or certification preparation.

Purpose:
To identify weaknesses in the declared carbon claim before external review begins.

What GeoCarb™ provides:

- early identification of potential claim overstatement risk;
- current-stage plausibility screening for ARR, restoration, agroforestry, and related projects;
- diagnostic routing for avoided-emission and conservation-type claims;
- clarification of whether a claim is directly screenable, partially screenable, or requires specialized review;
- guidance on where MRV resources should be focused.

04. Certification Preparation Teams

Who:
Consultants, project owners, and technical teams preparing nature-based projects for certification pathways.

Use case:
Pre-registration plausibility screening before formal validation or verification processes begin.

Purpose:
To assess whether the carbon claim entering the certification preparation process has already been screened for physical plausibility, claim type, and boundary coherence.

What GeoCarb™ provides:

- independent pre-certification screening reference;
- claim-type and boundary classification;

- screening-adjusted result for internal review;
- identification of assumptions that may require additional evidence before formal validation.

GeoCarb™ does not replace certification standards. It helps improve the quality of the claim before the certification process begins.

05. Governments and Policy Institutions

Who:
National climate finance authorities, NDC coordinators, ministries, public investment agencies, and policy institutions.

Use case:
Portfolio-level screening of nature-based project pipelines across regions, project types, and carbon-claim structures.

Purpose:
To support more disciplined allocation of climate finance toward projects whose claimed carbon outcomes are physically plausible and structurally coherent.

What GeoCarb™ provides:

- pipeline-level screening of NbS carbon claims;
- comparability assessment across project types;
- identification of claims requiring further review;
- portfolio-level signal of where carbon claims appear plausible, uncertain, or materially divergent;
- country-context risk interpretation within ENHANCE's protected methodology.

How to Engage

A. Public Screening Report

Purpose:
High-level screening output for early review, public-facing interpretation, or initial inquiry.

Includes:

- project context;
- claim comparability status;
- screening result;
- PASS / WARNING / DISTORTION flag;
- high-level interpretation;
- non-certification disclaimer.

Request:
contact@enhance-institute.org

B. Institutional Screening Report

Purpose:
Expanded diagnostic interpretation for institutions, funders, buyers, project owners, or portfolio reviewers under controlled engagement.

Includes:

- deeper claim-structure interpretation;
- expanded diagnostic reasoning;
- review priorities;
- MRV focus areas;
- institution-facing screening notes;
- protected internal logic withheld from public disclosure.

Not publicly disclosed:

- calibration parameters;
- routing thresholds;
- sensitivity ranges;
- protected diagnostic rules;
- internal risk modifiers;
- ENHANCE model-integration logic.

Request:
contact@enhance-institute.org

C. Pipeline Assessment

Purpose:
Batch screening for NbS project portfolios, carbon-credit pipelines, or national/regional project sets.

Use cases:

- portfolio due diligence;
- climate finance pipeline review;
- pre-submission project screening;
- buyer-side carbon claim review;
- country or regional NbS project comparison.

Website:
enhance-institute.org/geocarb

Public Disclosure Note

GeoCarb™ v2.0 outputs are screening signals only.

They are not certified carbon credits, issued credit volumes, registry-approved results, legal determinations, investment recommendations, or substitutes for project-level MRV.

GeoCarb™ is designed to help users ask the right question before a carbon claim becomes capital:

What kind of carbon process is being claimed — and is that claim physically defensible within a declared screening boundary?

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contact@enhance-institute.org | enhance-institute.org

Citation Hur, Junyoung. 2026. Carbon Claim Screening for REDD+ and NbS: A Physical Plausibility and Comparability Screen for NbS Carbon Claims Before Funding, Purchase, or Certification. ENHANCE Institute. Public Research Report. June 2026. Document Code: ENH-GC-RPT-001-2026.

Model Reference Hur, Junyoung. 2026. GeoCarb™: A Latitude-Based Carbon Absorption Model for Independent Pre-Verification of Nature-Based Solution Projects — Species-Specific Gaussian Calibration and Application to Climate Finance Due Diligence. SSRN Working Paper. April 13, 2026.

Model Version GeoCarb™ v2.0

Benchmarking Basis NASA MODIS MOD17A3 NPP · 27 Latitude Bands · 7 Vegetation Classes · R^2 = 0.91 Field Case Signal A: TIST Kenya Agroforestry Programme · VCS-verified programme context · Exeter University independent assessment · Core benchmark signal: Consistent (+3.8%) Field Case Signal B: Laos illustrative REDD+ screening context · Protected REDD+ diagnostic pathway · Screening-adjusted result 1,478,405 tCO_2e/yr · DISTORTION (–59.7%)

Risk Context Country-context risk screening available within ENHANCE's protected methodology.

Classification Publicly Disclosed — ENHANCE Institute, June 2026
Document Code ENH-GC-RPT-001-2026

Carbon claims should be classified before they are compared, and compared before they become capital. Classification before comparison. Diagnosis before capital.

The Gravity of Truth

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ENHANCE Institute is the research division of ENHANCE Co., Ltd. It develops and applies deterministic structural evaluation frameworks for climate projects and climate-finance environments.

ENHANCE's diagnostic architecture operates across the climate-project lifecycle. C-FAIR™ evaluates the structural financing environment of countries before proposals are designed. PathFinder-GOV™ translates country-level baselines into pre-submission project direction guidance for verified LDC/SIDS National Focal Points. VERA-TCV™ — comprising VERA™, ToC:f(x)™, C-FAIR™, and VORTA™ — provides integrated ex-ante structural audit of completed project design documents. ESF™ functions as ENHANCE's proprietary scenario-context operating field across all diagnostic layers.

These frameworks are designed to make structural conditions readable before they become implementation outcomes.

ENHANCE Institute publishes public research reports disclosing the findings of its diagnostic systems. It works with researchers, climate-finance institutions, accredited entities, and government focal points seeking rigorous structural evaluation before capital is committed.